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Jeremy A. Collins

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Website || Google Scholar || LinkedIn

EDUCATION

Georgia Institute of Technology **Aug 2023 – Present**
Ph.D. Machine Learning *Atlanta, GA*

Georgia Institute of Technology **Aug 2021 – May 2023**
M.S. Robotics, GPA: 4.00 *Atlanta, GA*

University of Arkansas **Aug 2017 – May 2021**
B.S. Mechanical Engineering, Minor in Mathematics, GPA: 4.00 *Fayetteville, AR*

- First-Ranked Senior Scholar, Honors College Fellow, Governor's Distinguished Scholar

RESEARCH EXPERIENCE

People, AI, and Robots (PAIR) Lab **Aug 2023 – Present**
Advised by Prof. Animesh Garg *Atlanta, GA*

- Scaling robot learning by harnessing diverse sources of data
- Enabling robots to learn from internet-scale video and perform dexterous manipulation

Healthcare Robotics Lab **Oct 2021 – Aug 2023**
Advised by Prof. Charlie Kemp *Atlanta, GA*

- Led creation of a transformer-based robotic policy that predicts force-based subgoals given a text prompt
- Devised deep learning methods to visually estimate contact in collaboration with Meta Reality Labs

Medical and Soft Robotics Lab **Sep 2019 – Jun 2021**
Advised by Prof. Yue Chen *Fayetteville, AR*

- Designed berry picking robotic gripper and novel peristaltic pump

PUBLICATIONS

- **AMPLIFY: Actionless Motion Priors for Robot Learning from Videos**
  Jeremy A Collins*, Loránd Cheng*, Kunal Aneja, Albert Wilcox, Benjamin Joffe, Animesh Garg
  In submission. Paper
- **COBALT: Crowdsourcing Robot Learning via Cloud-Based Teleoperation with Smartphones**
  Ayush Agarwal*, Ansh Gandhi*, Jeremy A Collins, Omar Rayyan, Aryan Sarswat, Ranjani Koushik, Kaavya Menon, Masoud Moghani, Ajay Mandlekar, Animesh Garg
  In submission. Paper
- **FLASH: Flow-Based Language-Annotated Grasp Synthesis for Dexterous Hands**
  Hrishit Leen, Jeremy A Collins, Kunal Aneja, Nhi Nguyen, Priyadarshini Tamilselvan Sri Siddarth Chakaravarthy P, Animesh Garg
  Conference on Robot Learning (CoRL) 2025 Workshop on Dexterous Manipulation. Paper
- **RoCoDA: Counterfactual Data Augmentation for Data-Efficient Robot Learning from Demonstrations**
  Ezra Ameperosa*, Jeremy A Collins*, Mrinal Jain, Animesh Garg
  IEEE International Conference on Robotics and Automation (ICRA) 2025. Paper
- **DexMOTS: Dexterous Manipulation with Differentiable Simulation**
  Krishnan Srinivasan, Jeremy A Collins, Eric Heiden, Ian Ng, Jeannette Bohg, Animesh Garg
  International Symposium of Robotics Research (ISRR) 2024. Paper
- **ForceSight: Text-Guided Mobile Manipulation with Visual-Force Goals**
  Jeremy A Collins*, Cody Houff*, You Liang Tan*, Charles C Kemp
  IEEE International Conference on Robotics and Automation (ICRA) 2024, CoRL 2023 demo. Paper
- **PressureVision++: Estimating Fingertip Pressure from Diverse RGB Images**
  Patrick Grady, Jeremy A Collins, Christopher D Twigg, Chengcheng Tang, James Hays, Charles C Kemp
  IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024, CVPR 2023 demo. Paper

- **Visual Contact Pressure Estimation for Grippers in the Wild**
  Jeremy A Collins, Cody Houff, Patrick Grady, Charles C Kemp
  IEEE/RSJ International Conf. on Intelligent Robots and Systems (IROS) 2023, IROS 2023 demo. Paper
- **Force/Torque Sensing for Soft Grippers using an External Camera**
  Jeremy A Collins, Patrick Grady, Charles C Kemp
  IEEE International Conference on Robotics and Automation (ICRA) 2023. Paper
- **Visual Pressure Estimation and Control for Soft Robotic Grippers**
  Patrick Grady, Jeremy A Collins, Samarth Brahmbhatt, Christopher D Twigg, Chengcheng Tang, James Hays, Charles C Kemp
  IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2022. Paper
- **Tendon-Driven Soft Robotic Gripper for Blackberry Harvesting**
  Anthony L Gunderman, Jeremy A Collins, Andrea Myers, Renee Threlfall, Yue Chen
  IEEE Robotics and Automation Letters (RA-L) 2022. Paper, Patent

INDUSTRY EXPERIENCE

Toyota Research Institute **May 2025 – Present**
Intern, Large Behavior Models Team
- Training foundation models for robotics
- Designing algorithms to leverage human videos for scaling pre-training

EleutherAI **May 2023 – Aug 2023**
Volunteer Researcher
- Explored text-conditioned image generation with limited access to text-image pairs

Dorabot **Jul 2021 – May 2022**
Engineering Intern
- Curated datasets for instance segmentation algorithms
- Designed and simulated robotic sorting systems to demonstrate ROI to clients such as DHL

Marshalltown Company **May 2019 – Aug 2019**
Mechanical Engineering Intern
- Improved robotic welding cell, increasing production speed of $3M in products annually
- Characterized manufacturing defect on high-priority product, increasing production by 75%

J.B. Hunt Transport Services **May 2018 – Aug 2018**
Application Development Intern
- Optimized codebase of flagship marketplace software (Shipper 360) to meet user specifications
- Added unit tests and increased the efficiency of the platform

DaVoice **May 2015 – Aug 2017**
Quality Control and Logistics
- Directed the sale and shipping of products through Amazon throughout high school

SKILLS

- **Programming Languages:** Python, C, C++, Arduino, MATLAB, Java
- **Tools:** PyTorch, TensorFlow, NumPy, Einops, SageMaker, OpenCV, Pandas, ROS, Isaac Gym, Git
- **Misc. Skills:** Competitive Rubik's Cube solver, trumpet player, amateur chocolatier

INVITED TALKS

- **AMPLIFY: Actionless Motion Priors for Robot Learning from Videos**
  Gemini Robotics Team, Google DeepMind **Jul 2025**
  RoboPapers Podcast **Aug 2025**
- **Causality, Consciousness, & Video Learning for Robotics**
  Georgia Tech AI Podcast **Mar 2025**

TEACHING

Georgia Institute of Technology

- Head Teaching Assistant, CS 8803 Deep Reinforcement Learning **Aug 2025 – Present**
- Teaching Assistant, CS 6601 Artificial Intelligence **Aug 2023 – May 2025**

SELECTED PROJECTS

Robot Learning from Human Feedback **Mar 2023 – Apr 2023**

- Trained a robotic policy by optimizing for human preferences using OpenAI gym
- Crowdsourced data collection by obtaining preferences from video pairs using Amazon Mechanical Turk

Video Prediction using Latent Diffusion **Sep 2022 – Dec 2022**

- Used an autoregressive transformer to predict future frame embeddings from Stable Diffusion encoder
- Predicted embeddings are denoised and decoded to produce future video frames

Learning Robotic Tasks from Video Observation **Sep 2022 – Dec 2022**

- Designed robotic system to learn behaviors using video data of expert demonstrations in OpenAI Gym

Multi-Agent Reinforcement Learning **Feb 2022 – May 2022**

- Implemented RL algorithms (SAC, PPO) using TensorFlow to explore human-robot interaction
- Simulated complex interactions between humans and robots using OpenAI Gym

HONORS/AWARDS

- Coauthor of funded proposal ($30k) from Georgia Tech Research Institute **May 2025**
- Coauthor of funded proposal ($60k) from AI-CARING **Oct 2022**
- Southern US Champion – Rubik's Cube Fewest Moves Challenge **Jun 2019**
- Best Poster, 2018 FEP Honors Innovation Symposium (Track 6) **Apr 2018**
- Presented independent research at the Joint Mathematics Meetings (JMM) funded by MAA **Jan 2018**
- 1st place, Fall 2017 J.B. Hunt Hackathon **Nov 2017**

COMMUNITY INVOLVEMENT

- President, Georgia Tech Robotics Graduate Student Organization **May 2024 – Present**
- Reviewer, ICRA 2023, NeurIPS 2024, CoRL 2024, ICRA 2024, CoRL 2025 **Nov 2023 – Present**
- Member, AI Safety Initiative at Georgia Tech **Sep 2022 – Present**
- Mentor and judge, HackMIT 2022-2023, HackGT 2022 **Oct 2022 – Sep 2023**
- President and founder, UArk Speedsolving Club (Rubik's Cube Club) **Aug 2017 – May 2021**
- Vice President, University of Arkansas Mathematics Club **Aug 2019 – Dec 2019**